
AN EDGE VISION IoT–LLM FRAMEWORK INTEGRATING MULTIMODAL SENSING AND BOUNDED LLM FEEDBACK FOR CONTINUOUS ERGONOMIC RISK SCREENING DURING COMPUTER USE *

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ABSTRACT

Prolonged computer usage among students and office workers is associated with musculoskeletal discomfort, particularly in the cervical and upper-back regions, and with the visual symptoms grouped under Computer Vision Syndrome (CVS). Existing ergonomic monitoring systems are commonly limited by single-factor analysis—focusing on posture or viewing distance alone—or depend on body-worn sensors and instrumented furniture. Equally important, systems that do combine sensing with AI-generated feedback generally stop at producing advice: they neither manage how often a user should be interrupted nor record whether the advice was acted upon. This paper presents **PostureCare**, an Edge Vision IoT–LLM framework built on a **Freenove ESP32-S3** microcontroller with an **OV3660** image sensor, an ultrasonic rangefinder, and a photoresistor. **MediaPipe** and **OpenCV** provide 3D head pose (Pitch, Yaw, Roll), Eye Aspect Ratio (EAR) and blink rate, while the physical sensors supply eye-to-screen distance and workspace illuminance. Rather than acting on instantaneous threshold crossings, detections are qualified by a persistence layer and classified by a four-tier severity model, and a **notification governor** converts qualified states into a bounded number of interventions using flag-set deduplication, exponential backoff, escalation, and user-controlled snooze and do-not-disturb. A cloud-based **Large Language Model** translates the fused state into a short, context-aware corrective recommendation delivered through a real-time web dashboard, synthesised speech, and a device-level indicator, operating within bounds set by the deterministic layers rather than deciding on its own when the user should be interrupted. This paper reports the architecture, the qualification and severity model, the governor mechanism, and the realised implementation, together with a full evaluation protocol. That protocol has not been executed: no measurement of detection accuracy, sensor calibration, interruption load, or user outcome is reported

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here. The contribution is accordingly a design and its implementation, not a validated clinical or behavioural result.

Keywords Computer Vision Syndrome · Ergonomics · ESP32-S3 · IoT · Large Language Models · Musculoskeletal Disorders · Multimodal Sensing · Notification Management

1 Introduction

Computers and digital display devices have become widely used in learning, office work, and remote working, resulting in prolonged screen exposure among students and employees [1, 2]. Prolonged computer use has been associated with both visual symptoms and work-related musculoskeletal disorders, with the neck, shoulders, upper back, and lower back among the body regions most frequently affected [1, 2]. Factors such as extended computer-use duration, static sitting, and inappropriate working posture have also been identified as relevant contributors to musculoskeletal problems among computer users [2]. In addition, deficiencies in workstation arrangement, monitor position, and lighting conditions may coexist with both musculoskeletal symptoms and visual discomfort [3]. These findings suggest that computer-use ergonomics should not be evaluated based on a single factor alone.

Among posture-related factors, head angle (HA) and viewing distance (VD) are two particularly relevant indicators. Hashimoto et al. reported that HA and VD were significantly associated with sagittal spinal alignment in computer users, especially in the cervical C2–C7 region [4]. Their findings also showed that a more forward spinal alignment tended to be accompanied by a larger head angle and a shorter viewing distance [4]. Earlier, Rempel et al. demonstrated that changes in display distance could lead to changes in head and trunk position during computer tasks [5]. Similarly, a study involving 309 Japanese office workers found that an inappropriate eye-to-monitor distance, particularly below 40 cm, was associated with greater neck pain intensity [6]. These findings support the use of head angle and viewing distance as quantitative indicators of postural ergonomics.

In addition to posture, ambient lighting is an important component of visual ergonomics. Visual conditions at computer workstations are influenced by factors such as illuminance, luminance, lighting direction, glare, correlated color temperature, and display characteristics [7]. Experimental evidence has also shown that changes in ambient illumination can affect visual fatigue during screen-based tasks [8]. However, a low-cost IoT system cannot assess all aspects of lighting quality. Therefore, the present study uses ambient light level as a simple environmental indicator that can be measured directly using a light sensor, rather than as a complete measure of lighting quality.

Another potentially useful visual indicator is blink rate. Chidi-Egboka et al. reported that blink rate decreased significantly during reading tasks, including tasks performed on digital devices, compared with non-reading activities [9]. More importantly, Lapa et al. demonstrated that blink rate could be detected and monitored using a conventional computer camera under real-life conditions; in their exploratory study, lower blink rates were associated with higher Computer Vision Syndrome scores [10]. However, blink rate can also vary according to task characteristics and cognitive demand [11]. Therefore, the present study treats blink rate as a supporting visual ergonomic indicator, rather than as an independent diagnostic measure.

Taken together, head angle, viewing distance, ambient light level, and blink rate represent different but complementary aspects of computer-use ergonomics. Head angle and viewing distance mainly reflect postural ergonomics, whereas blink rate and ambient light level provide additional information related to visual ergonomics. In addition, exposure duration should also be considered because an unfavorable posture or condition lasting only a few seconds is not equivalent to the same condition being sustained for a prolonged period [2, 12]. A multimodal assessment that incorporates temporal information may therefore provide a more complete representation of the user's ergonomic condition than a single-threshold warning.

Several technologies have been developed to automate ergonomic monitoring. Wearable-sensor systems can provide continuous motion-related information, but they require users to attach and maintain additional devices on the body [13]. For example, ERG-AI used data from five tri-axial accelerometers together with machine learning and large language models to predict occupational postures and generate health-related recommendations [14]. Another approach is to integrate sensors into furniture. SitWell used two 32×32 pressure-sensing mats embedded in a chair to classify 19 sitting postures, monitor posture duration, and generate personalized feedback using GPT-4o [15]. Although these systems demonstrate the potential of combining sensing and AI, they still depend on wearable devices or specialized hardware [14, 15].

Computer vision provides a non-contact and potentially lower-cost alternative. RGB cameras combined with pose-estimation models can be used to estimate head, shoulder, and upper-body posture [16]. However, the accuracy of computer-vision-based ergonomic assessment depends on factors such as camera position and the selected pose-estimation model [16]. Cameras can also be used to monitor visual behavior such as blink rate [10]. This creates the

possibility of using a single camera for both postural monitoring and visual-behavior monitoring, without requiring separate eye-tracking or wearable devices.

Although the individual components of the problem have been investigated, current studies generally examine posture, viewing distance, lighting, blink behavior, and AI-based feedback in separate research contexts or systems [4, 7, 10, 14, 15, 16]. Recent intelligent ergonomic systems have begun to move from simple detection toward personalized recommendations [14, 15], but they commonly rely on wearable sensors or smart furniture. Therefore, there remains limited research on a compact, low-cost, non-contact IoT-AI system that simultaneously integrates head angle, viewing distance, blink rate, ambient light level, and exposure duration, and then uses the combined information to generate context-aware recommendations.

A second gap sits behind the first and receives less attention. Continuous monitoring produces far more detections than a user can usefully act upon, so a system must also decide *which* detections warrant interrupting the user, how often the same condition may be raised again, and whether the resulting advice was acted upon. Existing intelligent ergonomic systems generate recommendations but do not manage this: they report detection performance and stop there, without reporting how many interruptions a user receives or whether behaviour changed [14, 15]. This matters in practice because repeated generic warnings that neither explain the cause nor identify what to adjust are the failure mode that leads users to disable such tools altogether — at which point measurement accuracy ceases to matter. Converting continuous multimodal detection into a bounded, measurable set of interventions is therefore a distinct problem from sensing it accurately.

This gap is important because the same detected posture may require different responses depending on the surrounding context. For example, a user with a large head angle but an appropriate viewing distance and a short exposure duration may not require the same level of intervention as a user who simultaneously maintains a large head angle, sits too close to the screen, works under unsuitable lighting, and remains in that condition for a prolonged period. Therefore, rather than simply classifying posture as correct or incorrect, a multimodal system may provide a more meaningful interpretation of why the condition is unfavorable and how it should be corrected.

To bridge these limitations, this research introduces an Edge Vision IoT-LLM framework, a cost-effective, multimodal system centered on the **Freenove ESP32-S3** microcontroller and the **OV3660** CMOS image sensor. The system leverages **MediaPipe** and **OpenCV** algorithms to perform real-time 3D head pose estimation (Pitch, Yaw, and Roll) alongside blink rate detection via Eye Aspect Ratio (EAR) analysis. Simultaneously, an ultrasonic sensor and a Light Dependent Resistor (LDR) provide precise telemetry for viewing distance and ambient illuminance, respectively. By synthesizing these data streams with cumulative exposure duration, the system conceptualizes the user’s ergonomic well-being through two complementary dimensions:

1. **Postural Ergonomic State:** Evaluated based on head pose orientation (Pitch, Yaw, Roll), eye-to-screen distance, and postural persistence.
2. **Visual Ergonomic State:** Evaluated based on ocular blink frequency (EAR-derived), workstation illuminance levels, and continuous screen-time duration.

These two dimensions organise the sensing and the interface. The intervention decision, however, is taken over the combined set: a user can act on only one correction at a time, so raising a postural and a visual alert simultaneously would put two instructions in competition for the same attention. Severity is therefore assigned once, over both dimensions together, as described in Section 3.4.

The integrated multimodal state is processed by a **Cloud-based Large Language Model (LLM)** reasoning engine to generate personalized, context-aware corrective recommendations, operating within bounds set by a deterministic notification layer rather than deciding on its own when the user should be interrupted. Vision processing runs on a client attached to the edge node rather than as an embedded network on the microcontroller, which permits a 468-landmark face mesh at the cost of transmitting frames off the device — a trade-off examined in Section 7.

It is important to note that the proposed framework is designed as a preventive screening and behavioral support tool; it is not intended for the clinical diagnosis of musculoskeletal disorders, Computer Vision Syndrome (CVS), or chronic ocular diseases. The primary objective of this study is to develop a low-latency, non-intrusive IoT-AI system that fuses postural, spatial, and environmental metrics into bounded, measurable real-time feedback. This paper reports the design and implementation of that system. The evaluation specified in Section 5 has not been carried out, and no claim of measured accuracy, interruption load, or behavioural effectiveness is made anywhere in what follows.

1.1 Research Questions

The questions below are *design* questions, answered by the architecture and its implementation. The empirical questions — how accurately the system measures, how well it discriminates ergonomic conditions, and whether its recommendations change behaviour — require a human-subjects study, and are stated as such in Section 5. They are outside the scope of what this paper reports.

- **RQ1:** What sensing arrangement allows postural, ocular, environmental, and temporal indicators to be captured concurrently on low-cost, non-contact hardware and fused into a single ergonomic state?
- **RQ2:** How can instantaneous detections be qualified so that transient movements are distinguished from sustained ergonomic exposure, and so that co-occurring conditions are treated differently from isolated ones?
- **RQ3:** How can the resulting qualified states be converted into a bounded number of user interventions, rather than one alert per detection?
- **RQ4:** How can a large language model be integrated as a bounded explanation layer, without granting it control over detection or over the decision to interrupt?

1.2 Expected Contributions

The main contributions of this study are summarized as follows:

1. A low-cost, non-contact multimodal ergonomic monitoring architecture based on an ESP32-S3 camera node combined with distance and ambient light sensors.
2. A comprehensive monitoring approach that integrates postural, visual, environmental, and temporal indicators rather than relying on a single variable, with two independent channels for viewing distance that cross-check one another.
3. An evaluation framework for jointly considering postural ergonomics and visual ergonomics in a unified system.
4. A mechanism for generating context-aware personalized corrective recommendations using AI, moving beyond simple single-threshold warnings.
5. A session-scoped acquisition state machine that ties measurement to detected desk presence, so that cumulative exposure pauses rather than resets when the user steps away — the condition under which exposure duration is a well-defined variable at all.
6. A persistence-gated notification governor that converts threshold crossings into bounded interventions through hold-time qualification, severity tiering, flag-set deduplication, exponential backoff, and user-controlled suppression.
7. An objective intervention-outcome measure in which each delivered reminder opens a fixed observation window recording whether the triggering condition cleared, yielding a corrected-per-reminder ratio per session.

2 Literature review

Computer-use ergonomics involves interactions among work duration, visual conditions, workstation arrangement, posture, and individual behavior. Previous monitoring systems commonly examine only one or two of these factors [14, 15, 16, 17]. This review therefore organizes the relevant literature by research theme and identifies the limitations addressed by the proposed **PostureCare** system.

2.1 Computer Use and Musculoskeletal Problems

Prolonged computer work has consistently been associated with discomfort in the neck, shoulders, upper back, and lower back [2]. Demissie et al. identified the lower back, neck, upper back, and shoulders as the body regions most frequently affected among computer users [2]. Reported risk factors included prolonged computer use, awkward posture, repetitive movement, and remaining in the same position for extended periods [2]. Workstation characteristics have also been associated with neck-pain intensity among office workers, indicating that computer-related discomfort depends not only on total exposure but also on how the workstation is arranged [6].

Duration is particularly important because computer work frequently involves sustained, low-variation postures [18]. A meta-analysis of 25 studies involving 43,184 participants found an association between sedentary behavior and neck

pain (OR = 1.46). Computer use was independently associated with greater neck-pain risk (OR = 1.23), while the association became stronger when sedentary duration reached at least four hours per day (OR = 1.60) or six hours per day (OR = 1.88) [12]. These findings support treating cumulative duration as part of ergonomic exposure rather than evaluating posture from a single image.

Static posture may maintain tension in the cervical and upper-back musculature even when the posture is only moderately different from neutral [18]. Repeated or prolonged exposure can consequently be more meaningful than a brief postural deviation [2, 12]. Continuous monitoring could identify increasing exposure and encourage corrective action before occasional discomfort becomes persistent. However, much of the existing evidence is observational and relies on self-reported computer use and symptoms; it establishes population-level associations but does not support individual clinical prediction [2, 6, 12].

2.2 Viewing Distance and Visual Ergonomics

Digital eye strain is influenced by viewing distance, screen position, text size, task demand, glare, blinking behavior, and uncorrected visual conditions [1, 11]. Viewing distance should therefore not be treated as an isolated value or as having one universally optimal threshold. A user who cannot comfortably read screen content may compensate by leaning forward, flexing the neck, or moving the entire trunk toward the display [5].

Rempel et al. examined 24 participants during two-hour computer tasks at mean viewing distances of 52.4, 73.0, and 85.3 cm [5]. When the monitor was placed at the farthest distance while character size remained unchanged, participants moved their heads and torsos closer to the display [5]. The authors recommended a distance between approximately 52 and 73 cm when character size is close to the user's visual-acuity limit [5]. This result demonstrates that a greater distance is not automatically more ergonomic: distance, text readability, and compensatory posture must be interpreted together.

Hashimoto et al. further demonstrated the relationship between viewing behavior and cervical alignment [4]. Using three-dimensional motion analysis in 21 participants, the researchers found that head angle and visual distance were strongly associated with C_2-C_7 and C_7-T_3 sagittal alignment. The respective correlation coefficients reached 0.8861 and 0.8460 when head angle and viewing distance were used together, whereas correlations with thoracic and lumbar alignment were substantially lower [4]. These results support including an independent distance sensor rather than estimating distance solely from face size in a camera image.

Visual behavior provides related but distinct information. Chidi-Egboka et al. found that blink rate decreased during reading from both printed and digital materials, regardless of viewing distance [9]. Lapa et al. also demonstrated that webcam-based blink monitoring was feasible in real-life computer-use settings and reported an association between lower blink rate and higher computer-vision-syndrome scores [10]. Blink rate may therefore indicate visual task demand, but it cannot substitute for direct distance measurement [9, 10]. The principal limitation of previous work is that viewing distance is often measured only during controlled experiments or inferred indirectly; few posture-monitoring systems continuously combine direct distance measurement with posture and exposure duration [4, 5, 17, 19].

2.3 Lighting and Workstation Environment

Lighting at a computer workstation involves more than ambient illuminance. Relevant factors include screen luminance, contrast between the screen and its surroundings, glare, reflections, light-source direction, surface reflectance, and the visual characteristics of the user [7]. Osterhaus et al. emphasized that these factors interact and should be considered together when evaluating visual comfort during computer work [7]. Experimental evidence has also demonstrated that luminance and illuminance conditions can affect visual fatigue and arousal during digital reading [20].

Lighting may also influence posture indirectly. In a repeated laboratory experiment involving 43 participants, Mork et al. found that exposure to direct glare produced greater forward bending of the head and changes in trapezius muscle blood flow during computer work [21]. This provides a plausible pathway in which an unfavorable visual environment produces squinting or repositioning, which subsequently changes head and upper-body loading [21].

For the proposed system, ambient-light measurements should be interpreted as contextual information. For example, poor posture detected under very low ambient illumination may lead to a recommendation to improve lighting and screen readability, whereas the same posture under adequate lighting may require adjustment of viewing distance or screen height. Nevertheless, a single illuminance sensor cannot directly measure glare, reflections, screen luminance, or individual visual ability; lighting should therefore remain a contextual ergonomic variable rather than a diagnostic indicator of musculoskeletal risk [7].

2.4 Sitting Posture and Ergonomic Risk

Postural factors relevant to computer use include forward-head position, excessive neck flexion, trunk flexion, slouching, lateral trunk lean, and shoulder asymmetry [4, 17, 19, 22]. Head angle and viewing distance can provide useful information about cervical alignment [4], but the relationship between visible posture and pain is not deterministic. For example, Sarig Bahat et al. did not find a clinically applicable association between forward-head posture and pain intensity or disability in a sample of participants with and without nonspecific neck pain [22]. Postural measurements should consequently be treated as exposure indicators rather than proof of an existing disorder.

The variables that can be measured depend strongly on camera position. A front-facing camera installed near the monitor can reliably observe frontal-plane relationships but has limited information about sagittal depth [16].

Table 1: Feasibility and Recommended Interpretation of Measurable Postural Variables

Posture Variable	Front-Camera Feasibility	Recommended Interpretation
Left-right shoulder-height difference	High	Shoulder-asymmetry indicator
Lateral head tilt or roll	High	Deviation from an individual neutral baseline
Head rotation or yaw	Moderate	Detect sustained rotation away from the screen
Lateral trunk lean	Moderate to high	Upper-body asymmetry indicator (if torso is visible)
Head position relative to shoulder midpoint	Moderate	Detect lateral displacement and large vertical changes
Forward-head displacement	Limited	Use only as a calibrated proxy
Neck flexion or head pitch	Limited to moderate	Estimate from facial orientation, not as a true cervical angle
Trunk flexion or slouching	Limited	Use shoulder and face displacement as indirect indicators

Personal calibration can improve interpretation by recording the user’s neutral face and shoulder positions before monitoring begins [19]. The external distance sensor can also distinguish actual movement toward the screen from apparent changes caused by image scale. The main limitation is that a front-facing monocular camera cannot directly measure sagittal cervical and trunk angles; it can monitor visible postural proxies but cannot diagnose forward-head posture, spinal deformity, or musculoskeletal disease [16].

2.5 Computer Vision-Based Posture Detection

RGB-camera monitoring offers a noncontact and comparatively low-cost approach to posture assessment [16, 19, 23, 24]. OpenPose detects two-dimensional body, hand, facial, and foot landmarks and supports multi-person pose estimation [23]. MediaPipe BlazePose is designed for lighter on-device inference and estimates 33 body landmarks for a single person, making it more suitable for mobile or edge-oriented applications [24]. Both approaches convert RGB images into anatomical landmark coordinates from which joint angles, alignment ratios, and posture classes can be calculated [23, 24].

PoseX demonstrated a computer-work-specific application using an ordinary webcam and OpenPose landmarks [19]. The system used proportional relationships among face width, shoulder width, and ear-to-shoulder height to recognize normal posture, cervical kyphosis, thoracic kyphosis, and raised shoulders. In an experiment involving 15 participants at viewing distances of 60, 85, and 110 cm, the system achieved at least 84% classification accuracy after a short personal calibration [19]. This supports the feasibility of personalized webcam-based posture classification but also demonstrates the dependence of image features on viewing distance.

Camera placement can materially affect measurement accuracy [16]. Murugan et al. compared MediaPipe BlazePose, VideoPose3D, 3D-pose-baseline, and PSTMO against reference measurements. The side-camera position produced the lowest mean absolute error across static and dynamic tasks, while the VideoPose3D full-body configuration performed best among the evaluated models [16]. Validation should therefore include multiple distances, lighting conditions, participant body sizes, clothing conditions, and partial occlusion. Appropriate outcomes include classification accuracy, precision, recall, F1-score, angular mean absolute error, and agreement with manually annotated or motion-capture measurements.

Camera-based monitoring is feasible, but most reported systems concentrate on posture classification alone [16, 17, 19]. Their principal limitations are small controlled datasets, sensitivity to camera position and occlusion, limited depth

information, and the infrequent integration of independently measured distance, lighting, and cumulative exposure [16, 17, 19].

2.6 Sensor-Based Ergonomic Monitoring

Nonvisual sensors can complement the limitations of monocular cameras. A distance sensor provides direct proximity measurement without relying on face size or camera calibration. A light sensor records changes in the workstation environment, while a timer converts individual posture events into cumulative exposure measures. IMUs can measure orientation and motion without depending on lighting, although they require attachment to the body and may experience calibration drift [13].

Sen et al. developed ERG-AI using 2,913 hours of data collected from 114 home-care workers wearing five tri-axial accelerometers. The system predicted occupational postures and estimated prediction uncertainty. Reducing the number of sensors improved wearability and potential energy efficiency but increased uncertainty and reduced the ability to distinguish more complex movements [14]. This illustrates the tradeoff between sensing completeness and practical deployment.

Pressure-based systems offer another alternative. Odesola et al. integrated two 32×32 pressure-sensor mats into an office chair and trained models to distinguish 19 sitting postures. Their convolutional neural network achieved 98.29% accuracy, and the resulting SitWell platform provided posture-duration analytics and personalized feedback [15]. However, the model was tailored using data from one individual, and the sensing hardware was tied to a particular chair.

Compared with multi-IMU systems, motion-capture equipment, or high-resolution pressure mats, a low-cost combination of RGB imaging, distance sensing, ambient-light sensing, and time logging is less intrusive and easier to install at an ordinary workstation. The central limitation is that low-cost sensor fusion requires careful calibration, synchronization, missing-data handling, and validation against reliable reference measurements before its combined risk indicators can be trusted.

2.7 AI-Based Ergonomic Feedback

The simplest feedback systems use fixed rules, such as issuing an alert when the viewing distance falls below a threshold or when poor posture continues for a specified duration. Rule-based feedback is computationally efficient and transparent, but repeated generic warnings may not explain the source of the problem or tell the user which workstation component should be adjusted. Experimental posture-correction systems have shown that real-time feedback can improve neck and trunk posture and reduce unnecessary muscle activity during computer work [25].

More adaptive systems incorporate personal baselines, historical exposure, and contextual measurements. ERG-AI transformed sensor-derived posture predictions and their uncertainty estimates into prompts for GPT-4 and LLaMA-based models. An ergonomics professional judged the resulting assessments to be generally meaningful and balanced, although the recommendations required greater specificity and additional personal information [14]. SitWell similarly used GPT-4o to generate recommendations from an individual’s historical posture data [15].

The value of an LLM in PostureCare is therefore not to replace sensor analysis or diagnose a disorder. Its role is to translate verified measurements into an understandable explanation. Instead of stating only “poor posture detected,” the system could explain that the user has remained closer than their calibrated viewing range while leaning toward the display and suggest increasing text size, moving the screen, or returning to the calibrated sitting position. Recommendations should be generated from structured sensor summaries and restricted to approved ergonomic actions.

Current LLM-based ergonomic systems remain limited by insufficient user studies, uncertain long-term behavior change, possible unsupported recommendations, and little clinical-outcome validation; the proposed LLM must therefore operate as a bounded explanation and recommendation layer rather than an autonomous medical decision-maker [14, 15, 17].

2.8 Literature Review Summary and Positioning of the Proposed Study

Previous studies demonstrate the feasibility of monitoring computer-use ergonomics through RGB-camera posture detection [16, 19, 23, 24], wearable sensors [13, 14], pressure-sensing furniture [15], distance measurement [4, 5, 6], and posture-correction feedback [14, 15, 25]. Nevertheless, these approaches commonly concentrate on only one or two dimensions of ergonomic exposure. Camera systems are generally posture-centric, wearable systems require body-mounted devices, and smart-chair systems depend on specialized furniture.

PostureCare is positioned as an integrated early ergonomic-risk monitoring and feedback system. The RGB camera provides observable frontal-posture indicators, the distance sensor independently measures eye-to-screen proximity, the

Table 2: Comparison of Existing Ergonomic Monitoring Approaches and the Proposed PostureCare System

Approach	Sensing and feedback					Intervention management		
	Post.	Dist.	Light	Dur.	AI fb.	Persist.	Notif.	Outcome
Camera posture system	Yes	Usually no	No	Sometimes	Limited	No	No	No
Wearable system	Yes	Usually no	No	Often	Limited	No	No	No
Distance-only monitoring	No	Yes	No	Sometimes	No	No	No	No
Smart chair / pressure sensing	Yes	No	No	Often	Sometimes	No	No	No
Proposed PostureCare	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Persist.: a detection must hold for a minimum duration before it is acted upon. *Notif.*: deduplication, backoff, escalation, and user-controlled suppression. *Outcome*: the system records whether a delivered recommendation was followed by a correction.

light sensor records environmental context, and time-series processing quantifies cumulative exposure. The LLM then converts these verified measurements into context-specific explanations and practical adjustment suggestions.

Two gaps therefore remain open, and the proposed system addresses both. The first is integration: no reviewed system combines postural, ocular, environmental and temporal channels on a single non-contact node. The second, visible in the rightmost columns of Table 2, is intervention management. Every approach reviewed here acts on a detection as soon as it occurs, reports detection performance, and stops. None requires a detection to persist before acting on it, none bounds how often the same condition may interrupt the user, and none records whether a delivered recommendation was followed by a correction. Since repeated undifferentiated warnings are what lead users to disable ergonomic software, converting continuous detection into a bounded and measurable set of interventions is a distinct problem from sensing accurately, and it is where this work concentrates.

The intended contribution is the integration and interpretation of several low-cost measurements, together with the mechanism that governs how those measurements reach the user, rather than the invention of a new clinical diagnostic method. The proposed system addresses the limited integration of posture, viewing distance, visual behavior, lighting context, and exposure duration identified in previous work [4, 5, 6, 7, 8, 9, 10, 12, 14, 15, 16, 17, 19, 20, 21, 22, 23, 24, 25].

3 Proposed System Architecture and Methodology

3.1 System Overview

PostureCare is organised as four stages, illustrated in Figure 1. Each stage narrows the data: raw frames and sensor samples become continuous signals, signals become discrete flags, flags that persist become a severity state, and only a bounded subset of severity states becomes an actual interruption of the user.

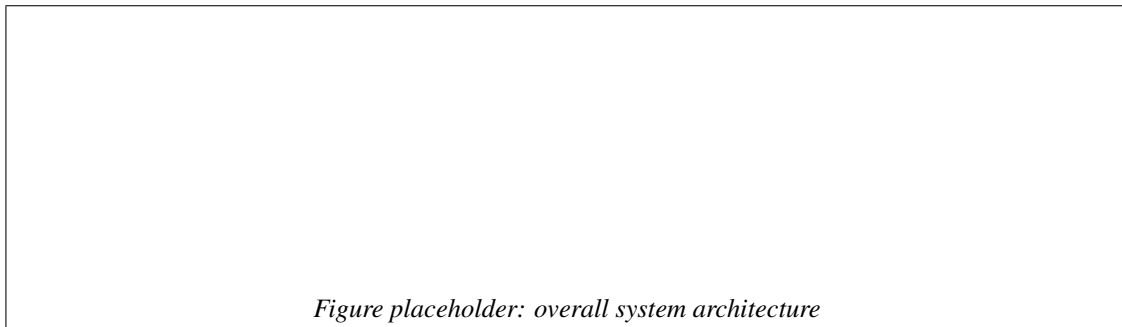


Figure 1: Overall system architecture. Stage 1 acquires frames and physical telemetry, Stage 2 synchronises them through Firebase, Stage 3 performs qualification and reasoning, and Stage 4 delivers bounded feedback through the dashboard, speech, and a device indicator.

Stage 1 – Edge acquisition. The Freenove ESP32-S3 captures the video stream from the OV3660 sensor and samples the ultrasonic rangefinder and photoresistor, publishing raw values to a Firebase Realtime Database.

Stage 2 – Vision processing and fusion. A client-side pipeline consumes the video stream and derives head orientation, Eye Aspect Ratio, blink rate, and a camera-based distance estimate. These are merged with the physical telemetry into a single ergonomic state.

Stage 3 – Qualification and reasoning. The fused state is reduced to a set of named flags; flags are qualified by how long they persist, classified into a severity tier, and, when a qualified state changes, passed to a cloud LLM for a short context-aware explanation.

Stage 4 – Bounded intervention. A notification governor decides whether the user is interrupted at all, through which channel, and how soon the same condition may be raised again.

It should be noted that image data is processed off the microcontroller rather than by an embedded neural network. This is a deliberate trade-off: it allows the use of a 468-landmark face mesh that would not fit the device’s compute budget, at the cost of transmitting frames off the device, which carries the privacy implications discussed in Section 7.

3.2 Hardware and Sensing

3.2.1 Edge Acquisition Module (Freenove ESP32-S3 & OV3660)

The edge sensing node is powered by the Freenove ESP32-S3 WROOM module equipped with an integrated 8 MB PSRAM and Wi-Fi connectivity. Image streams are captured using an OmniVision OV3660 3-Megapixel camera module [26]. The module is mounted at eye level at the top-centre of the primary monitor; the photoresistor is placed flat on the work surface so that it samples the light falling on the task area rather than the light reaching the screen bezel.

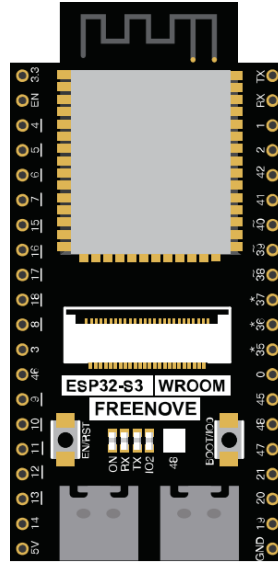


Figure 2: Freenove ESP32-S3 Board with Integrated Camera Module

The camera is streamed at QVGA (320×240) resolution to keep the wireless link within budget. This is well below the sensor’s native capability and is reported here because the accuracy of landmark-derived quantities depends on it. In particular, the camera-based distance estimate (Section 3.3) uses a scale factor expressed in pixel units and is therefore resolution-dependent: changing the capture source invalidates the calibration, which is why calibration is a mandatory step rather than an optional refinement.

3.2.2 Multimodal Physical Telemetry Sensors

- **Eye-to-Screen Distance Telemetry:** An Ultrasonic Sensor (HC-SR04) mounted adjacent to the monitor measures real-time distance (Z_{ultra} in cm) between the user and the workstation screen [27].
- **Workspace Illuminance Monitoring:** A Light Dependent Resistor (LDR) module collects ambient workspace illuminance levels (L_{ambient} in Lux) [28].

Distance is therefore observed through *two independent channels*: the ultrasonic rangefinder and the camera-based estimate. This is a cross-validation arrangement rather than redundancy. The ultrasonic sensor provides a scale reference

that does not depend on image geometry, and remains available when no face is detected; the camera estimate provides continuity when the ultrasonic returns no echo. The two do not measure the same point — the rangefinder returns the nearest surface within its cone, typically the face or chest plane, whereas the camera estimate is referenced to the interpupillary line — so their agreement bounds precision rather than establishing accuracy.

The illuminance channel converts the raw 12-bit ADC reading to lux by inverting the voltage divider to the photoresistor’s own resistance and applying the power-law characteristic of a CdS cell, which is a straight line in log–log space. Two reference points fully determine that line. The coefficients used in this work are nominal rather than measured against a photometric reference, and the consequences are stated in Section 7.

3.3 Computer Vision and Ocular Telemetry Pipeline

Instead of on-device embedded neural networks, the vision pipeline leverages a client-side computer vision stream built on **MediaPipe** [29] for spatial posture estimation and ocular fatigue analysis.

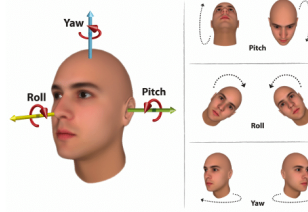


Figure 3: Facial Coordinate Frame Definition for Yaw (θ_y), Pitch (θ_p), and Roll (θ_r). Positive pitch corresponds to the head rotating downward toward the desk.

3.3.1 3D Head Pose Estimation via MediaPipe Face Mesh

1. **Landmark Extraction:** MediaPipe Face Mesh extracts 468 3D facial keypoints from the camera feed in real time [29].
2. **Perspective-n-Point (PnP) Alignment:** Selected 2D landmark coordinates (nose tip, chin, outer eye corners, mouth corners) are projected onto a standard 3D facial model using OpenCV’s `solvePnP` algorithm [30].
3. **Euler Angle Derivation:** The resulting rotation matrix yields the three canonical head orientation angles: Pitch (θ_p), Yaw (θ_y), and Roll (θ_r).

The reported angles are expressed *relative to a per-user zero reference* captured during a short calibration step at the start of each session, not relative to an anatomical axis. A value of $\theta_p = 0$ therefore means “the orientation this user adopted when asked to sit neutrally”, not “the cervical spine is in neutral alignment”. This choice follows the personal-calibration approach reported for webcam posture systems [19] and is what makes the angles comparable within a session; its consequences for interpretation are addressed in Section 7.

3.3.2 Ocular Fatigue & Blink Frequency Analysis (EAR)

Visual fatigue is evaluated by computing the Eye Aspect Ratio (EAR) using six facial landmarks per eye [31]:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \quad (1)$$

A blink event is logged when EAR falls below a threshold of 0.294 for three consecutive frames. This operating point differs from values commonly quoted in the literature because the magnitude of Equation 1 depends on which landmark indices form the ratio; the present pipeline uses a specific six-point subset per eye, and the threshold was tuned empirically for that subset rather than inherited. Blink events are accumulated over a *sliding 60-second window* to yield the real-time blink rate in blinks per minute, with a warm-up correction of $n/t \times 60$ during the first minute of a session so that the reported rate reflects current behaviour rather than a cumulative average.

3.4 From Signals to Qualified Ergonomic States

The fused state is expressed as the feature vector $X = [P, D, L, T]$, where P denotes postural orientation, D viewing distance, L ambient illuminance, and T cumulative exposure. Rather than mapping X to a single mutually exclusive

posture label, the system evaluates each signal independently and produces a *set* of named flags. This matters because the situations the system is intended to distinguish are precisely those in which several factors coincide: sitting close to the screen in a dimly lit room is a different ergonomic condition from either factor alone, and an exclusive classifier discards exactly that information.

Table 3: Flag vocabulary. Flags in the danger set D qualify after a shorter hold time and are sufficient on their own to raise the alert tier.

Flag	Set	Condition
head_too_low	danger	$\theta_p > +5^\circ$ (head rotated down toward the desk)
head_too_high	danger	$\theta_p < -5^\circ$
head_tilted	danger	$ \theta_r > 15^\circ$
critically_close	danger	Viewing distance below 30 cm
head_turned	notice	$ \theta_y > 20^\circ$
too_close	notice	Viewing distance below 50 cm
too_far	notice	Viewing distance above 70 cm
too_dark	notice	Illuminance below 300 lx
too_bright	notice	Illuminance above 500 lx
low_blink_rate	notice	Blink rate below 6 min^{-1}

Let $F(t)$ be the set of flags whose condition holds at time t :

$$F(t) = \{ f : \text{condition}_f \text{ holds at } t \} \quad (2)$$

A threshold crossing alone is not treated as an ergonomic exposure. Glancing down at the keyboard produces a momentary crossing that is not equivalent to the same posture sustained for minutes, a distinction the epidemiological evidence on static loading makes explicitly [2, 12, 18]. Each flag must therefore *persist* for a minimum hold time τ_f before it qualifies:

$$Q(t) = \{ f \in F(t) : \text{hold}_f(t) \geq \tau_f \} \quad (3)$$

where $\text{hold}_f(t)$ is the elapsed time since flag f last became continuously true, and τ_f is 15 s for the danger set and 60 s otherwise. The qualified set $Q(t)$ is then mapped to a severity tier:

$$S(t) = \begin{cases} \text{escalated} & \text{if the alert tier has held for } \geq 300 \text{ s} \\ \text{alert} & \text{else if } Q \cap D \neq \emptyset \text{ or } \text{too_close} \in Q \text{ or } |Q| \geq 2 \\ \text{notice} & \text{else if } |Q| = 1 \\ \text{normal} & \text{otherwise} \end{cases} \quad (4)$$

The $|Q| \geq 2$ clause is what operationalises the multimodal argument: two individually mild conditions occurring together are treated as more serious than either alone, without requiring a trained model to say so.

Postural variables not represented in Table 3 were excluded deliberately. Forward-head displacement, trunk flexion, and slouching are rated of limited feasibility for a front-facing monocular camera in Table 1, and including them would have meant reporting quantities the sensing arrangement cannot support.

3.5 Session Lifecycle and Exposure Measurement

Cumulative exposure is only a meaningful variable if it measures time the user actually spent at the desk. Monitoring is therefore scoped to an explicit session with the states *idle*, *calibrating*, *monitoring*, *away*, *break*, and *ended*. Three rules govern the exposure clock:

- Exposure accumulates **only** in the *monitoring* state with a face detected — not merely while the application is open.
- Loss of face detection for 30 s moves the session to *away* and **pauses** the clock. Returning within 180 s resumes the accumulated total; beyond that the exposure counter restarts. A naive timer would otherwise either reset on every dropped frame or continue counting an empty chair.

- A 20 s grace period after session start excludes settling-in samples, and those samples are marked so that they are also excluded from session-level aggregates.

An idle session closes automatically after 900 s without the user returning. The exposure measured under these rules is *continuous uninterrupted desk exposure*, which is the quantity relevant to break-taking guidance. It is not an estimate of daily cumulative sedentary load, and no claim is made that it maps onto the daily-duration odds ratios reported in Section 2.

3.6 Notification Governor

Detecting an unfavourable condition and interrupting the user are different decisions. A system that raises an alert on every qualified state produces the repetitive generic warnings criticised in Section 2, and the practical consequence is that users disable it. The governor sits between qualification and delivery and applies four rules:

- **Deduplication.** The invocation key is the qualified flag set itself. A condition that persists unchanged does not generate a new explanation; a change in the set does.
- **Exponential backoff.** If the same condition remains uncorrected, reminders are repeated on a lengthening schedule rather than at a fixed interval. The backoff resets once the state has been normal for 120 s, so a later, unrelated violation is still raised promptly.
- **Escalation.** An alert sustained for 300 s escalates and requires explicit acknowledgement, which is the design’s position on alert fatigue: rather than increasing frequency, the system increases salience once and then waits.
- **User control.** Snooze suppresses reminders for 600 s and a do-not-disturb mode suppresses them indefinitely, both reversible from the dashboard.

Governor state is persisted with the session record, so cooldowns survive a restart or a second open dashboard rather than resetting — without this, a page reload would silently re-enable an alert the user had just dismissed.

Each delivered reminder additionally opens a 60 s observation window and records whether the triggering flags cleared within it. This yields a corrected-per-reminder ratio per session and is the instrument for the behavioural question posed in Section 5.

3.7 Feedback and Intervention Channels

Interventions are delivered through three channels:

- **Device indication.** The ESP32-S3 drives a two-colour indicator from a state object in the Realtime Database: green while monitoring normally, red at the alert tier, and blinking red when escalated. Fetching and blinking are non-blocking so that camera streaming and sensor sampling are unaffected.
- **Speech.** A single-sentence spoken form of the recommendation is delivered through browser speech synthesis. It is opt-in and enabled during guided setup.
- **Interface.** A transient notification carries the governor’s available actions (snooze, do-not-disturb, acknowledge). A break suggestion is raised when continuous exposure reaches its threshold, and is synchronised bidirectionally with the built-in focus timer so that a timed break also suspends posture alerting.

3.8 Cloud LLM Integration and Ergonomic Reasoning

The role of the language model [32] is to translate a measured state into an explanation, not to detect, diagnose, or decide when to interrupt — those decisions are made by the deterministic layers above. Invocation is *event-driven*: a request is issued when the qualified flag set changes, not on a timer. The distinction is material to deployment cost. At the backend’s 10 s sampling interval, a periodic design would issue on the order of 360 requests per hour of sustained poor posture, whereas the governor issues one per distinct condition plus scheduled repetitions under backoff.

Two bounded prompt modes are used. In *advice* mode the model is constrained to the detected flags and asked to explain why that combination is unfavourable and what to adjust. In *explain* mode, used when the user requests an assessment while all readings are in target, the model is explicitly instructed not to invent a problem. Output follows a fixed Context/Advice/Spoken contract, the third field being the single short sentence used for speech.

When the API is unavailable the system falls back to a deterministic per-flag phrase table and labels the resulting record accordingly, so that non-generated text is distinguishable in the logs.

3.9 Data Schema and Operating Parameters

Table 4 lists the variables recorded for each sample, and Table 5 the operating parameters that govern how they are interpreted.

Table 4: Multimodal data variables and system roles

Variable	Type	Description / Ergonomic Significance
Session_ID	Categorical	Identifier scoping all rows to one monitoring session
Timestamp	Time	UNIX timestamp for multi-sensor stream synchronisation
State	Categorical	Session state (idle, calibrating, monitoring, away, break, ended)
$\theta_p, \theta_y, \theta_r$	Continuous	Head orientation Euler angles in degrees ($^\circ$), relative to the calibrated zero
EAR	Continuous	Instantaneous Eye Aspect Ratio
Blink Rate	Continuous	Blink frequency over a sliding 60-second window (min^{-1})
Distance (camera)	Continuous	Interpupillary-distance estimate of user-to-screen distance (cm)
Distance (ultrasonic)	Continuous	Independently measured user-to-screen distance (cm)
Light Reading	Continuous	Workspace ambient illuminance (Lux), with a range status flag
Face Present	Boolean	Whether a face was detected; gates exposure accumulation
Exposure	Continuous	Continuous uninterrupted desk exposure (s)
Flag Set	Set	Conditions currently true (Table 3)
Flag Durations	Continuous	Per-flag persistence used by Equation 3
Severity	Categorical	Tier assigned by Equation 4
Reminders / Corrected	Integer	Interventions delivered, and those followed by a correction

All operating parameters are read from a single configuration file, so that the values below are the values the running system uses. Table 5 lists them in full to support replication.

4 Implementation

This section describes the realised artifact: the components that implement the architecture of Section 3, and the technologies each is built on. It reports what exists, not how well it performs. **No integration testing across the complete pipeline and no evaluation have been carried out**; the protocol for the latter is given in Section 5, and the consequences for what may be claimed are stated in Section 7.

4.1 Component Structure

The system is implemented as the four deployable components listed in Table 6, communicating through a Firebase Realtime Database. The database decouples them: the edge node does not need to know whether a dashboard is connected, and the vision client and backend can be restarted independently of the hardware.

4.2 Design Choices Worth Recording

Three implementation decisions follow from the architecture rather than from convenience, and are recorded here because they constrain how the system can be deployed and extended.

A single configuration source. Every threshold, hold time, severity rule and governor parameter in Table 5 is read from one configuration file on each request, and the calibration values are editable from the dashboard at runtime. This exists so that the values described in this paper are demonstrably the values the system uses, and so that a deployment can be retuned for a different user or workspace without redeployment.

Table 5: Operating parameters as configured

Parameter	Value	Role
<i>Detection thresholds</i>		
Pitch	$\pm 5^\circ$	Head down / back from calibrated zero
Roll	$\pm 15^\circ$	Lateral head tilt
Yaw	$\pm 20^\circ$	Rotation away from the screen
Viewing distance	50–70 cm	Target range
Distance, critical	< 30 cm	Distinct acute band
Illuminance	300–500 lx	Workspace ambient target
Blink rate	$\geq 6 \text{ min}^{-1}$	Sliding 60 s window
EAR	0.294 / 3 frames	Blink event threshold and debounce
<i>Temporal qualification (τ_f)</i>		
Danger flags	15 s	Head-pose and critically-close hold
Notice flags	60 s	Lighting, blink rate, distance, head turned
Escalation	300 s	Sustained alert without correction
<i>Notification governor</i>		
Backoff	60, 180 s	Repeat schedule for an unchanged condition
Snooze	600 s	User-initiated suppression
Normal reset	120 s	Clear time before backoff returns to first step
Correction window	60 s	Observation window after a reminder
<i>Session</i>		
Grace	20 s	Post-start settling, excluded from aggregates
Face lost $\rightarrow$ away	30 s	Pause exposure accumulation
Away $\rightarrow$ reset	180 s	Beyond this, exposure restarts
Idle $\rightarrow$ end	900 s	Auto-close the session
Break suggestion	1200 s	Continuous exposure trigger
Sampling interval	10 s	Backend; vision pipeline runs at $\approx 6\text{--}7 \text{ Hz}$

Table 6: Implemented components and their technologies

Component	Technology	Responsibility
Edge firmware	C++ / Arduino, PlatformIO	Camera streaming server, ultrasonic and photoresistor sampling, two-colour indicator driven from the database, publication of raw telemetry
Vision client	Python, MediaPipe, OpenCV	Face mesh extraction, head pose by solvePnP, Eye Aspect Ratio and sliding-window blink rate, interpupillary distance estimate, per-flag persistence timers
Backend service	Python, FastAPI, SQLite	Session state machine, flag detection and qualification, severity assignment, notification governor, LLM invocation, session history
Dashboard	HTML, CSS, vanilla JavaScript	Live telemetry, guided calibration, notification surface with governor actions, calibration and threshold configuration

The governor is the only writer of user-facing state. Session persistence, indicator state, history, and LLM invocation are all driven from the session engine rather than from independent timers. Without this, a page reload or a second open dashboard would reset cooldowns and re-raise an alert the user had already dismissed — the failure mode that makes notification management ineffective in practice.

Deterministic fallback for the recommendation layer. When the language model is unreachable, a per-flag phrase table supplies the recommendation and the record is labelled with the fallback model name. This keeps the system usable offline, and it also means generated and templated output can later be compared directly on identical detected states, which is the basis of the comparison described in Section 5.

4.3 Illuminance Conversion

The photoresistor is sampled as a raw 12-bit value and converted to lux in the backend rather than on the device, so that the conversion has a single definition. Two properties of that conversion are worth stating because they bound what the illuminance channel can report.

First, the divider is inverted to the photoresistor’s own resistance in normalised form, in which the supply voltage cancels; supply voltage is therefore not a parameter, which removes a source of error given that the regulated rail is not exactly nominal.

Second, the conversion is bounded at both ends. Above the point at which the analogue-to-digital converter saturates, the photoresistor still has finite resistance, so a saturated sample means “at least this bright” rather than a specific value. The implementation returns a range status alongside the value and reports such samples as a lower bound. Treating saturation as zero illuminance — which a naïve divider inversion does, since it computes zero resistance at full scale — would invert the reading and raise a low-light warning under the brightest conditions the sensor can register.

4.4 Status

All components described in Section 3 are implemented, and the individual processing stages have been exercised during development. What has *not* been done is end-to-end integration testing of the complete pipeline under realistic conditions, and no measurement of accuracy, latency, interruption load, or user outcome is reported anywhere in this paper. Establishing those is the immediate next step and is specified in Section 5.

5 Planned Evaluation Protocol

The system is built to support the evaluation described here, but that evaluation has *not been carried out*. This section is written in the future tense throughout and states a protocol, not results. It is included because the design decisions in Section 3 were made against these measurements, and because specifying them makes the boundary between what has been demonstrated and what remains unvalidated explicit.

5.1 Design

The evaluation separates into the four studies listed in Table 7, which have different requirements. Combining them into a single protocol, as is common, would force sensor characterisation and behavioural observation into the same session even though the first needs no participants and the second needs unscripted work.

Table 7: Planned studies and what each answers

Study	Participants	Answers
A — Bench validation	none	Sensor accuracy and angular repeatability
B — Scripted conditions	8–10	Detection performance, condition discrimination
C — Free use	6–8, 2 sessions	Recommendation quality, correction behaviour
D — Replay ablation	none (uses B, C logs)	Contribution of temporal qualification

5.2 Study A: Bench Validation

Distance readings from both channels will be compared against a tape measure at fixed marks between 30 and 90 cm, reporting mean absolute error and root-mean-square error per channel and the agreement between channels. Illuminance will be compared against a calibrated reference meter across a range of lighting conditions; if no reference instrument is available, the claim reduces to repeatability and to agreement with intended lighting categories, which is weaker but honest.

Head orientation requires a different treatment. Because angles are expressed relative to a user-set zero (Section 3.3), there is no anatomical true value to be accurate against, and a mean absolute error against an anatomical reference would be meaningless. The measurable properties are **repeatability** — the spread of reported angle across repeated returns to the same marked head position — and **sensitivity** — whether a known angular displacement Δ produces a reported change of Δ . Both will be assessed against marked positions on a fixed rig.

5.3 Study B: Scripted Conditions

Participants will hold each of the six conditions in Table 8 for a fixed interval.

Table 8: Experimental conditions

Condition	Scenario	Description
1	Normal ergonomic condition	Appropriate posture, suitable distance, normal lighting
2	Close viewing distance	Normal posture, intentionally reduced user-screen distance
3	Forward head posture	Forward head position with otherwise normal conditions
4	Excessive neck flexion	Head tilted downward beyond the neutral range
5	Low lighting	Normal posture and distance under inadequate lighting
6	Combined risk condition	Forward head, close distance, and poor lighting together

Ground truth comes from the assigned condition and from manual annotation of recorded video, never from the system’s own output. Reported metrics will be per-flag precision, recall and F1-score with a confusion matrix, and, for blink detection, precision and recall against frame-level annotation of a ten-minute segment per participant.

Eight to ten participants is sufficient to populate a confusion matrix across six conditions with readable per-cell counts. It is not sufficient to claim generalisation across body size, eyewear, or skin tone, and that limitation will be stated rather than implied.

5.4 Study C: Free Use

Participants will work normally for a fixed period with the system active. Two measures follow.

Recommendation quality will be rated on five-point Likert scales for relevance, clarity, usefulness, and actionability, collected on a form and joined to the session record by session identifier. To test whether the language model earns its dependency, the generative and deterministic paths will be alternated within participant and counterbalanced, holding the detection layer identical. A within-subject design is used deliberately: at these sample sizes a between-subjects comparison has no realistic power.

Correction behaviour will be measured by the corrected-per-reminder ratio and time-to-correction per flag type, both already instrumented. The unit of analysis here is the **reminder, not the participant**: six to eight participants over two sessions each yields on the order of a hundred reminders, which is a usable denominator even though the participant count is small. Results will be reported descriptively; no claim of statistical significance will be made at this sample size.

5.5 Study D: Replay Ablation

Logs from Studies B and C will be replayed with the qualification layer disabled, isolating its contribution to interruption load on identical input. This requires no additional data collection and provides the comparison against a single-factor threshold baseline.

5.6 Ethics

The study involves continuous facial video and therefore requires informed consent covering: that frames are transmitted off the device for processing, how long recordings are retained and who can access them, that spoken audio prompts form part of the intervention, and the participant's right to end a session at any point. Since the system is a screening and behavioural-support tool rather than a diagnostic instrument, participants will be told explicitly that its output is not a health assessment.

6 Discussion

6.1 Summary

This work set out four design questions. The sensing arrangement of Section 3.2 captures postural, ocular, environmental and temporal indicators concurrently on a single non-contact node, with two independent channels for viewing distance (RQ1). Equations 2–4 separate transient movement from sustained exposure through a per-flag hold time, and raise the severity tier when conditions co-occur (RQ2). The governor of Section 3.6 converts qualified states into a bounded number of interventions through deduplication, backoff, escalation and user-controlled suppression (RQ3). The language model operates inside those bounds as an explanation layer, invoked only when the qualified flag set changes and constrained to the detected flags (RQ4).

These are design answers, supported by an implementation rather than by measurement. They say nothing about how accurately the system senses, how well it discriminates ergonomic conditions, or whether its recommendations change behaviour. All three remain open until the protocol in Section 5 is executed, and nothing in this paper should be read as evidence for them.

6.2 Comparison with Previous Studies

ERG-AI derives occupational posture from five body-worn accelerometers and generates health recommendations with a large language model [14]; SitWell classifies nineteen sitting postures from chair-embedded pressure mats and produces personalised feedback with GPT-4o [15]. Both demonstrate that sensing and language models can be combined productively, and both depend on hardware the user must wear or on a specific item of furniture. Camera-based systems such as PoseX avoid that dependency and reach useful classification accuracy after a short personal calibration [19], but remain posture-centric.

The present system differs in two respects. It combines postural, ocular, environmental and temporal channels on a single non-contact node, and it treats the decision of *whether to interrupt* as a distinct problem with its own mechanism. In the systems above, an assessment leads directly to feedback; here, an assessment must first persist, then survive deduplication and backoff, before it is delivered. No superiority claim is made: the comparison is one of scope, and the accuracy comparisons that would justify a stronger statement have not been performed.

6.3 Why Multimodal Monitoring Matters

The same posture carries different implications depending on its context. A user with a raised head angle at an appropriate viewing distance, sustained briefly, is not in the same ergonomic situation as a user who simultaneously holds that angle, sits closer than 30 cm, works under inadequate lighting, and remains so for twenty minutes. A single-threshold system reports the same alert in both cases.

Equation 4 operationalises this: two individually mild conditions occurring together raise the severity tier, which is a claim about combinations rather than about any single signal. Whether that distinction matters in practice — that is, what proportion of interventions would arise from combinations rather than from any single threshold — is measurable from session logs, and is one of the quantities the protocol in Section 5 is designed to obtain. It has not been measured here.

6.4 Detection Is Not Intervention

Continuous monitoring produces far more detections than a person can usefully act on. A system that surfaces each one converts a health tool into a source of interruption, and the predictable consequence is that the user disables it — at which point measurement accuracy no longer matters at all. This failure mode receives little attention in the ergonomic monitoring literature, which generally evaluates detection performance and stops there.

The mechanisms in Section 3.6 take a concrete position on it. Persistence filtering removes transient movements that are not exposures. Deduplication on the qualified flag set prevents an unchanged condition from generating repeated explanations. Backoff lengthens the interval when a reminder goes unheeded rather than repeating at fixed frequency, and escalation increases salience once, requiring acknowledgement, instead of increasing frequency indefinitely. Snooze and do-not-disturb give the user a proportionate response short of disabling the system entirely.

Whether this produces better adherence than a simpler scheme is an empirical question that has not been tested here.

6.5 Practical Implications

The hardware is inexpensive, non-contact, and mounts on an ordinary monitor without modifying the workstation, which suits student, home-office and shared-desk settings where instrumented chairs or body-worn sensors are impractical. Because all thresholds, hold times and governor parameters are read from a single configuration file and are adjustable from the dashboard, the same deployment can be retuned for a different user or workspace without redeployment — a practical requirement for any ergonomic tool intended to be used by more than one person.

7 Limitations

The system has not been evaluated. No human-subjects study was conducted, and no end-to-end integration testing of the complete pipeline has been carried out. Detection accuracy, sensor calibration, interruption load, latency, and the effectiveness of the intervention layer are all unmeasured. The thresholds, hold times and severity rules in Table 5 are design values chosen from published ergonomic guidance, not values validated against observation. This is the principal limitation, it bounds all the others, and it means the contribution of this paper is an architecture and its implementation rather than an empirical result.

The remaining limitations fall into three groups.

7.1 Measurement

- **Head angles are relative, not anatomical.** Orientation is measured against a user-set zero captured at calibration, so values express deviation from an individual’s chosen neutral rather than a cervical angle. The relationship between visible head posture and symptoms is in any case not deterministic: Sarig Bahat et al. found no clinically applicable association between forward-head posture and pain intensity [22]. Postural output should be read as an exposure indicator, never as evidence of a disorder.
- **A front-facing monocular camera cannot recover sagittal depth.** Forward-head displacement, trunk flexion and slouching are excluded for this reason (Table 1), so the postural picture is partial by construction.
- **The two distance channels reference different points.** Their agreement bounds precision; it does not establish accuracy for either.
- **Illuminance is derived, not measured.** The ADC-to-lux conversion is a modelled curve over a photoresistor divider whose coefficients are nominal rather than fitted against a photometric reference. Above the divider’s saturation point the reading is censored — reported as a lower bound rather than a value. With the fixed resistor used here, the trustworthy ceiling falls close to the lower edge of the target illuminance band, so ordinary desk lighting can push the converter beyond its linear region.
- **A single illuminance sensor cannot characterise lighting quality.** Glare, reflections, light direction, screen luminance and colour temperature all matter for visual comfort [7] and none are observed. Ambient light is a contextual variable only.
- **Blink rate is not an independent diagnostic.** It varies with task demand and cognitive load [11], and is used here as a supporting indicator.

7.2 Scope of Operation

- **Exposure is counted only while a face is detected**, so poor lighting or occlusion cause under-counting in exactly the conditions the system exists to flag.
- **Exposure is continuous, not cumulative**. The measured quantity is uninterrupted desk time within a session and does not estimate daily sedentary load, so it cannot be compared against the daily-duration odds ratios cited in Section 2.
- **Thresholds and hold times are derived from published guidance and from calibration, not from validation against ergonomic assessment by an expert**. They are plausible defaults, not established cut-points.
- **The recommendation layer degrades to fixed phrases** when the language model API is unreachable, and generated and templated output are not equivalent.
- **Camera placement, occlusion and lighting influence computer-vision performance** [16], and the streamed resolution is well below the sensor's capability.

7.3 Deployment

- **Image data leaves the device**. Vision processing is performed off the microcontroller, so any real deployment must address retention, access and transmission security for facial video — a requirement an on-device inference design would have avoided.
- **No long-term adherence data**. Whether the governor's design actually prevents the alert fatigue it is intended to prevent has not been observed over any meaningful period.
- **The system provides ergonomic risk screening and behavioural support**. It does not diagnose musculoskeletal disorders, Computer Vision Syndrome, or any other condition, and its recommendations have not been reviewed by ergonomics or healthcare professionals.

8 Conclusion

This paper presented PostureCare, an edge vision IoT-LLM framework for continuous ergonomic risk screening during computer use. The system combines a Freenove ESP32-S3 with an OV3660 camera, an ultrasonic rangefinder and a photoresistor, and fuses head orientation, blink rate, viewing distance, ambient illuminance and exposure duration into a single ergonomic state on low-cost, non-contact hardware.

Its distinguishing element is what happens after detection. Rather than acting on instantaneous threshold crossings, the system qualifies each detection by how long it persists, classifies the qualified set into a four-tier severity model in which co-occurring mild conditions outrank isolated ones, and passes the result to a notification governor that bounds how often the user is interrupted through deduplication, exponential backoff, escalation with acknowledgement, and user-controlled suppression. A cloud language model operates within those bounds as an explanation layer, and each delivered reminder opens an observation window recording whether the condition cleared.

The contribution is an architecture and its implementation. All components described here are built, and the design choices that constrain deployment — a single configuration source, a single writer for user-facing state, and a deterministic fallback for the recommendation layer — are recorded in Section 4. The system has not been evaluated: detection accuracy, sensor calibration, interruption load and behavioural effectiveness are all unmeasured, and a full protocol for establishing them is specified in Section 5.

Executing that protocol is the immediate next step. Beyond it, three directions follow from the present design: separating severity per ergonomic dimension so that postural and visual conditions can be reasoned about independently; adding daily cumulative exposure so that the temporal variable connects to the epidemiological literature this work currently holds at arm's length; and moving inference on-device, which would remove the privacy cost the current architecture accepts in exchange for a richer landmark model.

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